Notebook 2 — Chapter 2.2

Concentration Inequalities & Tail Bounds

20 worked examples — Markov, Chebyshev, Jensen, WLLN, CLT — calibrated for Stage 1 speed and Stage 2 trading intuition

A. Markov's Inequality

Example 1 — Basic Bound

A desk's daily trading volume X (in \$M, non-negative) has $\mathbb{E}[X] = 20$. Bound $\mathbb{P}(X \geq 100)$.

Solution. $\mathbb{P}(X \geq 100) \leq \dfrac{\mathbb{E}[X]}{100} = \dfrac{20}{100} = \boxed{0.20}$.

⋆ *Intuition:* Markov needs only non-negativity and the mean — no shape, no variance. That's its power (works anywhere) and its weakness (very loose, as Example 4 will show).

Example 2 — Markov on a Transformed Variable

Non-negative loss X has $\mathbb{E}[X] = 5$, $\mathbb{E}[X^2] = 34$ (so $\sigma^2 = 9$). Bound $\mathbb{P}(X \geq 20)$ two ways: using $\mathbb{E}[X]$ directly, and using $\mathbb{E}[X^2]$.

Solution. Direct: $\mathbb{P}(X \geq 20) \leq \frac{5}{20} = 0.25$. Via X^2 (valid since $X \geq 0 \Rightarrow X \geq 20 \iff X^2 \geq 400$): $\mathbb{P}(X^2 \geq 400) \leq \frac{34}{400} = \boxed{0.085}$.

▶ *Speed trick:* applying Markov to a power or exponential of X (not X itself) is the general move — it's exactly how Chebyshev (Section B) and Chernoff bounds are built. Always ask "what transform of X makes my known moment more informative?"

Example 3 — When Markov Is Exactly Tight

Let $X = 0$ w.p. 0.7 and $X = 10$ w.p. 0.3. Verify Markov's bound at $a = 10$ is achieved with equality.

Solution. $\mathbb{E}[X] = 0.3(10) = 3$. Markov: $\mathbb{P}(X \geq 10) \leq \frac{3}{10} = 0.3$. Actual: $\mathbb{P}(X \geq 10) = \mathbb{P}(X = 10) = 0.3$. **Equality.**

⋆ *Intuition:* a two-point distribution with all its mass exactly at 0 or exactly at a is the canonical case where Markov cannot be improved — any smoothing of this distribution loosens the bound.

Example 4 — Fixed Points, and How Loose Markov Can Be

For a random permutation of n cards, $\mathbb{E}$[fixed points] $= 1$ (Notebook 1). Bound $\mathbb{P}(\geq 5$ fixed points), and compare to the true value for $n = 52$.

Solution. Markov: $\mathbb{P}(\geq 5) \leq \frac{1}{5} = \boxed{0.20}$. True value (fixed points $\approx$ Pois(1) for large n): $\mathbb{P}(\geq 5) = 1 - \sum_{i=0}^{4} \frac{e^{-1}}{i!} \approx 0.0037$.

⋆ *Intuition:* the bound (0.20) overshoots the truth (≈ 0.004) by 50×. Markov is a *worst-case* guarantee over all distributions with the given mean — real distributions are almost always far better behaved. Contrast with Example 3's tightness.

Example 5 — Chernoff Preview: a Cautionary Result

$X \sim \text{Exp}(1)$ (so $\mathbb{E}[X] = 1$). Bound $\mathbb{P}(X \geq 3)$ via the optimized exponential (Chernoff) form $\mathbb{P}(X \geq a) \leq \min_{t>0} e^{-ta}\mathbb{E}[e^{tX}]$, and compare to plain Markov and the exact value.

Solution. MGF: $\mathbb{E}[e^{tX}] = \frac{1}{1-t}$, $t < 1$. Minimize $f(t) = e^{-ta}/(1-t)$: $f'(t) = 0 \Rightarrow t^* = 1 - \frac{1}{a} = \frac{2}{3}$. Bound $= e^{-2}/(1/3) = 3e^{-2} \approx \boxed{0.406}$.

Plain Markov: $\mathbb{E}[X]/3 = 1/3 \approx 0.333$. Exact: $\mathbb{P}(X \geq 3) = e^{-3} \approx 0.0498$.

⋆ *Intuition:* here plain Markov (0.333) actually **beats** the optimized Chernoff bound (0.406)! Exponential/Chernoff tilting shines for *sums of many* i.i.d. variables (tightening exponentially as n grows, used constantly in Stage 2 risk sizing) — for a single variable it offers no free lunch. Never assume "fancier bound = tighter bound" without checking.

B. Chebyshev's Inequality

Example 6 — Basic Two-Sided Bound

Daily P&L X has $\mu = \$10$k, $\sigma = \$4$k. Bound $\mathbb{P}(|X - 10| \geq 12)$ (in \$k).

Solution. $k = 12/4 = 3 \Rightarrow \mathbb{P}(|X - \mu| \geq 3\sigma) \leq \frac{1}{9} \approx \boxed{0.111}$.

Example 7 — Sample Size for a Confidence Target

Per-trade variance $\sigma^2 = 4$ (\$^{2}). How many independent trades n guarantee $\mathbb{P}(|\bar{X}_n - \mu| \geq 0.10) \leq 0.05$?

Solution. $\text{Var}(\bar{X}_n) = \sigma^2/n$, so $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$. Solve $\frac{4}{n(0.01)} = 0.05 \Rightarrow n = \frac{4}{0.0005} = \boxed{8000}$.

⋆ *Intuition:* 8,000 trades is a *huge* number for just ±\$0.10 accuracy at 95% — Chebyshev is distribution-free and therefore very conservative. Example 18's CLT-based approach will need far fewer.

Example 8 — One-Sided Chebyshev (Cantelli)

For $k > 0$: $\mathbb{P}(X - \mu \geq k\sigma) \leq \dfrac{1}{1+k^2}$. A risk desk only cares about upside loss exceeding 2σ. Compare Cantelli to naively halving the two-sided Chebyshev bound.

Solution. Cantelli ($k = 2$): $\leq \frac{1}{1+4} = \boxed{0.20}$. Naive halving of two-sided bound: $\frac{1}{2} \cdot \frac{1}{k^2} = \frac{1}{2} \cdot \frac{1}{4} = 0.125$.

⋆ *Intuition:* the "halving" trick is **invalid** without a symmetry assumption Chebyshev never makes — it can even understate risk, as it does here (0.125 ¡ true valid bound 0.20). Cantelli is the rigorous one-sided tool; derived by applying Markov to $(X - \mu + u\sigma)^2$ and optimizing over u.

Example 9 — Chebyshev Tightness Construction

Build a distribution with mean μ, variance σ^2, for which $\mathbb{P}(|X - \mu| \geq k\sigma)$ exactly equals $1/k^2$.

Solution. Let $X = \mu - k\sigma$ w.p. $\frac{1}{2k^2}$, $X = \mu + k\sigma$ w.p. $\frac{1}{2k^2}$, $X = \mu$ w.p. $1 - \frac{1}{k^2}$. Check: $\text{Var}(X) = (k\sigma)^2 \cdot \frac{1}{k^2} = \sigma^2$ ✓. $\mathbb{P}(|X - \mu| \geq k\sigma) = \frac{1}{2k^2} + \frac{1}{2k^2} = \frac{1}{k^2}$ exactly. At $k = 2$: $\boxed{0.25}$, matching the bound with equality.

Example 10 — Markov vs. Chebyshev: Applicability, Not Just Tightness

Desk P&L X has $\mu = \$5$k, $\sigma = \$2$k, and *can go negative*. Bound $\mathbb{P}(|X - 5| \geq 6)$.

Solution. Markov **cannot** be applied to X directly (P&L isn't non-negative). Chebyshev needs no such restriction: $k = 6/2 = 3 \Rightarrow \mathbb{P}(|X - 5| \geq 6) \leq \frac{1}{9} \approx \boxed{0.111}$.

⋆ *Intuition:* Chebyshev only needs a finite variance — it's the general-purpose tool whenever your random variable can be negative (returns, P&L, mark-to-market) where Markov simply doesn't apply.

C. Jensen's Inequality

Example 11 — Why Variance Is Never Negative

Use Jensen to show $\text{Var}(X) \geq 0$ always.

Solution. $g(x) = x^2$ is convex, so $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ by Jensen. Rearranging: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq \boxed{0}$, with equality iff X is a.s. constant.

Example 12 — Harmonic vs. Arithmetic Mean in Position Sizing

Position sizes (in \$M) are $2, 4, 6$ with equal probability. Compare $\mathbb{E}[1/X]$ to $1/\mathbb{E}[X]$.

Solution. $g(x) = 1/x$ is convex for $x > 0$: Jensen gives $\mathbb{E}[1/X] \geq 1/\mathbb{E}[X]$. $\mathbb{E}[X] = 4 \Rightarrow 1/\mathbb{E}[X] = 0.25$. $\mathbb{E}[1/X] = \frac{1}{3}\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6}\right) = \boxed{0.306}$. Indeed $0.306 > 0.25$.

⋆ *Intuition:* this is exactly why average cost-per-share (harmonic-mean-flavored, weighting small trades more) is never less than the naive reciprocal-of-average-size — a classic VWAP-calculation trap.

Example 13 — Option Convexity: Where Jensen's Gap Is the Option's Value

Stock price S is \$80 or \$120, each w.p. 0.5. Strike $K = \$100$, payoff $g(S) = \max(S - K, 0)$. Compare $\mathbb{E}[g(S)]$ to $g(\mathbb{E}[S])$.

Solution. $\mathbb{E}[S] = 100 \Rightarrow g(\mathbb{E}[S]) = \max(0, 0) = 0$. $\mathbb{E}[g(S)] = 0.5(0) + 0.5(20) = \boxed{\$10}$.

⋆ *Intuition:* the entire \$10 gap *is* the option's time/volatility value — the textbook reason convex payoffs are worth strictly more than their "payoff at the expected price." More spread in S (holding $\mathbb{E}[S]$ fixed) only widens this gap — the foundational intuition for why volatility has positive price.

Example 14 — Quantifying Jensen's Gap (Second-Order Approximation)

For g twice differentiable, Taylor expansion gives $\mathbb{E}[g(X)] - g(\mu) \approx \frac{1}{2}g''(\mu)\,\text{Var}(X)$. Apply to $g(x) = \sqrt{x}$ at $\mu = 100$, $\sigma^2 = 25$.

Solution. $g''(x) = -\frac{1}{4}x^{-3/2}$, so $g''(100) = -\frac{1}{4000}$.

$$\mathbb{E}[\sqrt{X}] - \sqrt{100} \approx \frac{1}{2}\left(-\frac{1}{4000}\right)(25) = \boxed{-0.0031}.$$

So $\mathbb{E}[\sqrt{X}] \approx 9.9969$ — slightly *below* $\sqrt{\mu} = 10$ (concave $g \Rightarrow$ negative gap, opposite direction from Example 13's convex case).

▶ *Speed trick:* the sign of g'' alone tells you the direction of Jensen's gap without computing anything — convex (options, x^2) $\Rightarrow$ gap ≥ 0; concave

($\sqrt{x}$, $\ln x$, utility functions) $\Rightarrow$ gap ≤ 0.

D. Weak Law of Large Numbers

Example 15 — Proving WLLN via Chebyshev

Fair-coin indicators $X_i \sim$ Bernoulli(0.5), $\sigma^2 = 0.25$. Using Chebyshev, bound $\mathbb{P}(|\hat{p}_n - 0.5| \geq 0.05)$ and find n making this bound ≤ 0.10.
Solution. $\text{Var}(\hat{p}_n) = \sigma^2/n = 0.25/n$. Chebyshev: $\mathbb{P}(|\hat{p}_n - 0.5| \geq 0.05) \leq \frac{0.25}{n(0.0025)} = \frac{100}{n}$. Set ≤ 0.10: $n \geq \boxed{1000}$.
$\star$ *Intuition:* this *is* the proof of WLLN — for any fixed $\varepsilon > 0$, $\sigma^2/(n\varepsilon^2) \to 0$ as $n \to \infty$, so $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \to 0$. Convergence in probability, nothing more, nothing less.

Example 16 — Applying WLLN at a Fixed, Large n

$n = 10{,}000$ independent trades, per-trade $\sigma = \$2$. Bound $\mathbb{P}(|\text{avg fill} - \text{fair value}| \geq \$0.05)$.
Solution. $\mathbb{P}(|\bar{X}_n - \mu| \geq 0.05) \leq \frac{\sigma^2}{n\varepsilon^2} = \frac{4}{10000(0.0025)} = \frac{4}{25} = \boxed{0.16}$.
$\star$ *Intuition:* even at 10,000 trades, Chebyshev only guarantees 84% confidence for a \$0.05 window — the price of being distribution-free. A CLT-based interval (Example 19) would be dramatically tighter for the same data.

Example 17 — The Trap: WLLN Is Not a "Correction" Guarantee

After 100 fair-coin flips, you observe 60 heads (10 more than expected). Does WLLN imply future flips will have *fewer* heads to "balance it out"?
Solution. **No.** WLLN says $S_n/n \to 1/2$ *in probability* as $n \to \infty$, driven by the denominator n growing, not by the numerator "correcting." In fact $\mathbb{E}[|S_n - n/2|]$ *grows* like $\Theta(\sqrt{n})$ in absolute terms — the raw excess-heads count does not shrink; only the *ratio* S_n/n converges.
$\star$ *Intuition:* this is precisely the Gambler's Fallacy, stated rigorously. Each future flip remains an independent fair coin — the past 10-head excess is diluted by a growing n, never erased. A classic Stage 1 trap: confusing convergence of a ratio with convergence (or correction) of a count.

E. Central Limit Theorem

Example 18 — De Moivre–Laplace with Continuity Correction

$n = 100$ fair coin flips, $S_{100} \sim$ approx $\mathcal{N}(50, 25)$. Find $\mathbb{P}(S_{100} \geq 60)$.
Solution. With continuity correction: $Z = \frac{59.5-50}{5} = 1.9 \Rightarrow \mathbb{P} \approx 1 - \Phi(1.9) \approx \boxed{0.0287}$. Without correction: $Z = 2.0 \Rightarrow \mathbb{P} \approx 0.0228$ (exact binomial value ≈ 0.0284).
▶ *Speed trick:* always apply the ± 0.5 continuity correction when using a continuous Normal to approximate a discrete Binomial/count — it alone closed most of the gap to the exact value here.

Example 19 — CLT for Portfolio P&L: Why Many Small Edges Compound

400 independent trades, each i.i.d. with mean \$50, sd \$200. Find $\mathbb{P}(\text{total P\&L} < 0)$.
Solution. $S_{400} \sim$ approx $\mathcal{N}(400 \cdot 50,\ 400 \cdot 200^2) = \mathcal{N}(20000,\ 16{,}000{,}000)$, $\sigma_{\text{tot}} = 4000$.

$$Z = \frac{0 - 20000}{4000} = -5 \ \Rightarrow\ \mathbb{P}(S_{400} < 0) \approx \boxed{2.87 \times 10^{-7}}.$$

$\star$ *Intuition:* this is the mathematical core of "why trading desks want many small, independent, positive-edge bets" — mean grows linearly in n while sd grows only as $\sqrt{n}$, so the Sharpe-like ratio $\mu\sqrt{n}/\sigma$ pushes the loss probability toward zero exponentially fast.

Example 20 — When CLT Needs Caution: Skewed, Small-n Bets

A trade pays +\$10 w.p. 0.99 or −\$990 w.p. 0.01 (mean \$0, break-even). For $n = 30$ such i.i.d. trades, is a Normal approximation to the sum trustworthy?
Solution. **No** — not yet. The Berry–Esseen theorem bounds the approximation error: $\sup_x |F_n(x) - \Phi(x)| \leq \frac{C\rho}{\sigma^3\sqrt{n}}$, where $\rho = \mathbb{E}[|X - \mu|^3]$.
Here ρ is huge relative to σ^3 because of the rare, extreme −\$990 outcome, so the bound stays large even at $n = 30$ — the true distribution is dominated by "did the rare bad event occur," not a bell curve.
$\star$ *Intuition:* this is the same fat-tail warning from Chapter 1.3's Normal section, now quantified — the more skewed/heavy-tailed the per-trade payoff, the larger n must be before CLT's bell-curve approximation is trustworthy. The "$n \geq 30$" rule of thumb silently assumes roughly symmetric, thin-tailed inputs.

Drill order: re-derive Examples 5, 8, 17, and 20 from scratch, closed-book — they're the four that overturn a naive first instinct (fancier bound isn't always tighter; you can't halve a two-sided bound; WLLN doesn't "correct" a streak; CLT needs enough n relative to skew). Those four carry this entire chapter's exam traps.